

a keyboard of all time

Problem ID: akeyboardofalltime

shhh.
wait.
did you hear that click?
they think I don't know what they're doing. But I see the them. I hear the keys at night. The clackclackclack like teeth in the walls.
i need to know. who types?? clack clack clack

the voices provide me with information about their keyboard and what they will type. BUT WHO TYPES WHAT???
the feds are after me, if i cant provide them this information they WILL put me in a squad car and place me in Guantanamo.
every step, one person can choose to make a move. this move is either to jump on a key to press it, or travel to another connected key. It is not possible for both people to make a move in the same step.

Input

The input consists of:

- One line containing two integers N ($1 \leq N \leq 26$) and M , the total number of keys and connected key pairs respectively.
- Then follow M lines. Each line contains two space-separated letters, a_i and b_i , representing two connected keys on the keyboard. Each key may either be a letter, or a sequence of numbers, the second case indicating a malfunctional key that serves no purpose. In either case, they are unique to the key.
- Then follows two space-separated letters, a_s and b_s , the starting key of person A and person B.
- Finally comes a line that contains the string W ($1 \leq |W| \leq 500$), the word that the people are going to type. This line is guaranteed to only include characters that are on the keyboard, and contains no white-space.

Output

Output a line containing a string of the same length as the word W , where each character is either 'A' or 'B', indicating which person typed each corresponding character in W . Then, output a second line containing an integer indicating the total number of steps taken to type the word.

Sample Input 1

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5 4
o f
f b
b a
b r
f b
foobar
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Sample Output 1

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AAABBB
10
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